

Yuting Zhou

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EDUCATION

Rutgers University , New Jersey, United States	Sep 2024 - Expected Jun 2026
• Major: M.S. Candidate in Computer Science	
Shanghai Institute of Technology , Shanghai, China	Sep 2020 - Jun 2024
• Major: Bachelor of Engineering in Software Engineering (GPA: 3.7, Top 10%)	

CERTIFICATES

- NVIDIA — Transformer-Based Natural Language Processing; Intro to Deep Learning
- Alibaba Cloud Certified Associate (ACA) — Big Data; Huawei Certified ICT Associate (HCIA) — Intelligent Computing

RESEARCH EXPERIENCE

Research Assistant <i>Hierarchical Text Classification, Data Engineering, Large Language Models</i>	Jun 2025 - Present
Rutgers Center for Advanced Infrastructure and Transportation (CAIT)	New Jersey, United States
• Architected scalable data engineering pipelines to merge and reconcile 100k+ multi-source railway incident and delay records, implementing automated deduplication, hierarchical label dictionary construction, and cross-dataset consistency validation.	
• Developed analytical and visualization tools to convert raw operational logs into structured daily, route-level, and location-based performance metrics, supporting delay pattern analysis and anomaly detection.	
• Designed and fine-tuned hierarchical multi-label classifiers based on pretrained transformer models (e.g., DeBERTa), modeling structured dependencies with supervised fine-tuning, encoder freezing strategies, and dependency-aware evaluation.	

INTERNSHIP EXPERIENCE

Front-end Engineer <i>JavaScript, Vue, React, WebSocket, MongoDB</i>	Jun 2023 - Aug 2023
Shanghai Tenchii Digital Tech Co., Ltd. (Tencent and Shanghai Metro Joint Venture)	Shanghai, China
• Optimized front-end performance of the Shanghai Transit App by refactoring component rendering and request handling logic, reducing page load latency by 30% and improving user retention.	
• Developed a real-time lottery system supporting high-concurrency traffic using WebSocket and MongoDB, ensuring low-latency updates and stable performance under peak load.	
• Designed and implemented 3 cross-team prototype interfaces in collaboration with product managers, translating Axure mockups into production-ready Vue components.	
Software Test Engineer <i>Pytest, JUnit, Selenium</i>	Jun 2022 - Sep 2022
Shanghai Spacecraft Electromechanical Equipment Co., Ltd.	Shanghai, China
• Built automated test environment across hardware configs with multi cases using Pytest/JUnit and CI pipeline.	
• Detected high-priority defects via Selenium UI automation, reducing production risk.	

PROJECT EXPERIENCE

Intelligent Rail Incident Classifier <i>Hierarchical Multi-label Learning, PLMs, LLMs, Large-scale Log Mining</i>	Aug 2025 - Present
• Designed and trained a hierarchical multi-label classifier based on DeBERTa-v3 to model structured dependencies on real-world railway incident logs.	
• Built an end-to-end training and inference pipeline with supervised fine-tuning, warm-up scheduling, encoder freezing, and reproducible experiment management.	
• Engineered scalable data preprocessing workflows for multi-source log merging, deduplication, hierarchical dictionary construction, and label consolidation across 3-year datasets.	
• Evaluated structured predictions using hierarchical exact match, per-level F1, and dependency-aware consistency checks.	
StreamX - Movies Recommender <i>Next.js, Django, ML, Pandas</i>  github.com/Magicherry/Movies-Recommendation	Feb 2026 - Present
• Engineered a custom Matrix Factorization model using SGD to optimize rating predictions and top-K recommendations.	
• Developed a full-stack Next.js and Django web app featuring multi-genre filtering and personalized user profiles.	
• Built an automated data pipeline with per-user 80/20 splits, evaluating models via MAE, RMSE, Precision@10, and NDCG@10.	
• Integrated TMDB API to dynamically enrich movie datasets with metadata, delivering a modern UI with glass-morphism effects.	
King of Bots <i>Vue 3, Spring Boot, MySQL, WebSocket, Redis</i>  github.com/Magicherry/King-of-Bots	May 2022 - Nov 2022
• Developed a full-stack real-time battle platform supporting 50+ concurrent matches with <250ms end-to-end latency.	
• Architected distributed session and matchmaking management using Redis caching and pub/sub to ensure state consistency under high concurrency.	
• Optimized game state synchronization and WebSocket multiplexing, increasing throughput by 35% under load testing.	
• Designed and implemented 10+ secure REST APIs with JWT authentication and role-based access control (RBAC).	

TECH STACK & LANGUAGES

Programming	Python, C++, Java, Go
ML & NLP	PyTorch, Transformers, scikit-learn, TensorFlow, DeBERTa / BERT, LLM Fine-tuning, Hierarchical Classification
ML Systems	Mixed Precision Training, Gradient Accumulation, Multi-GPU / MPS Training, Experiment Tracking
Data & Analysis	NumPy, Pandas, Matplotlib, Seaborn, Feature Engineering
Databases	SQLite, MySQL, Redis, MongoDB
Frontend/Backend	HTML, CSS, JavaScript, Vue 3, React, Next.js, Django, FastAPI, Spring Boot, RESTful APIs, Microservices
DevOps	Linux, Docker, Kubernetes, GitHub Actions
Languages	English, Mandarin Chinese (Bilingual)